

Project 2 Circuit Design 1

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2.A Part A

2.A.1 Objective

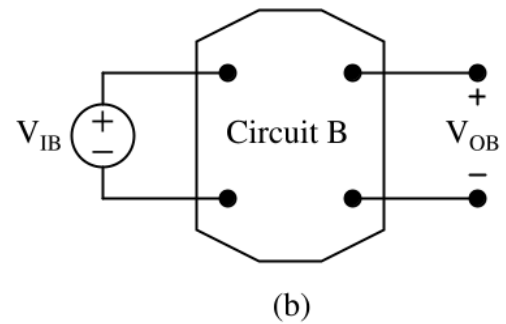
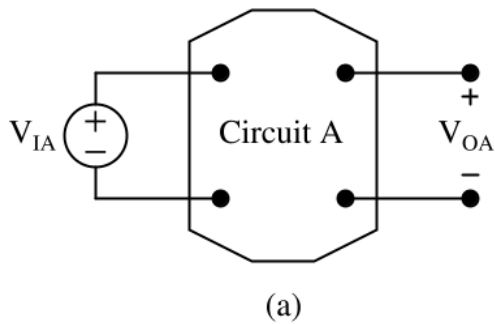


Figure 1: Circuit Design Part A

We were expected to design circuits A and B such that their output had a specified gain proportional to their inputs. This Requirement is represented by Equation 1 and Equation 2

$$\begin{aligned} V_{OA} &= K_A \times V_{IA} \\ V_{OB} &= K_B \times V_{IB} \end{aligned} \quad \text{Eq 1}$$

$$\begin{aligned} K_A &= 0.5 \\ K_B &= 0.9 \end{aligned} \quad \text{Eq 2}$$

2.A.2 Design

The obvious choice to satisfy the design requirement is to use a voltage divider. Using a simple divider consisting of two resistors in the $1k\Omega$ to $10k\Omega$ range also satisfies both design constraints. The formula (Equation 3) for this simple voltage divider (Figure 2) tells us that the ratio of the first resistor to the total resistance defines the gain.

Placeholder

Figure 2: Simple Voltage Divider

$$V_{\text{Out}} = V_{\text{In}} \times \frac{R_1}{R_1 + R_2} \quad \text{Eq 3}$$

In order to satisfy the design requirements, we will choose $1k\Omega$ resistors for R_1 in both circuits. We will then choose $1k\Omega$ and $9k\Omega$ for R_2 in circuits A and B respectively. The resulting circuits are shown in Figure 3 and Figure 4.

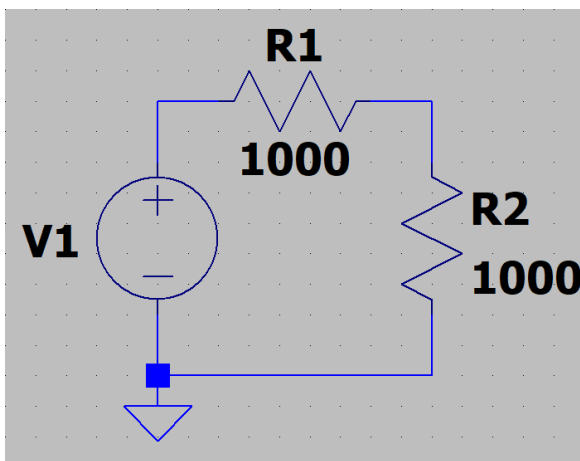


Figure 3: Circuit A

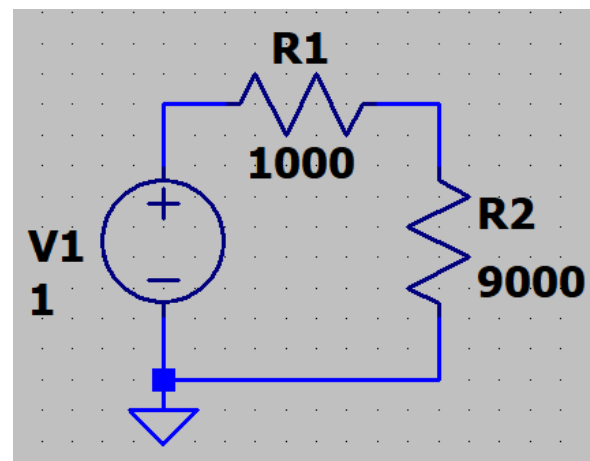


Figure 4: Circuit B

2.A.3 Analysis Results

2.A.3.a Circuit A

KCL@ V_{Out}

$$\frac{V_{\text{In}} - V_{\text{Out}}}{R_1} = \frac{V_{\text{Out}}}{R_2} \quad \text{Eq 4}$$

$$\frac{V_{\text{In}} - V_{\text{Out}}}{1k\Omega} = \frac{V_{\text{Out}}}{1k\Omega}$$

```
import sympy as sp
Vi, Vo = sp.symbols('Vi Vo')
eq1 = (Vi - Vo)/1000 - Vo/1000
print(sp.solve(eq1, Vo))
```

```
>> [Vi/2]
```

$$\therefore V_{\text{Out}} = 0.5 \times V_{\text{In}} \quad \text{Eq 5}$$

2.A.3.b Circuit B

KCL@ V_{Out}

$$\frac{V_{\text{In}} - V_{\text{Out}}}{R_1} = \frac{V_{\text{Out}}}{R_2} \quad \text{Eq 6}$$

$$\frac{V_{\text{In}} - V_{\text{Out}}}{1k\Omega} = \frac{V_{\text{Out}}}{9k\Omega}$$

```
import sympy as sp
Vi, Vo = sp.symbols('Vi Vo')
eq1 = (Vi - Vo)/1000 - Vo/9000
print(sp.solve(eq1, Vo))
```

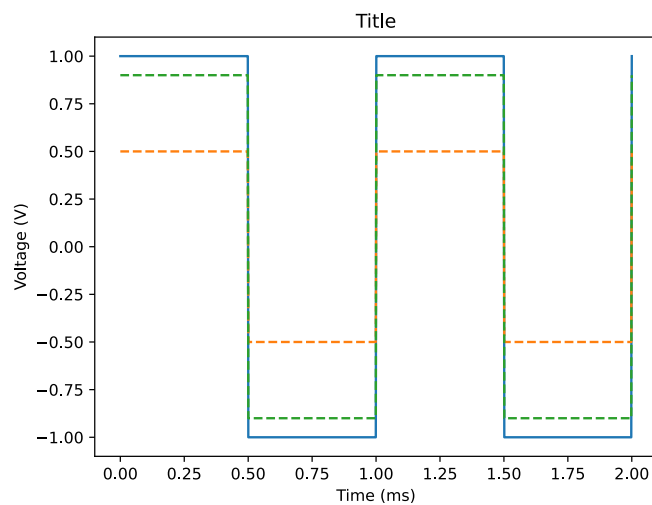
```
>> [9*Vi/10]
```

$$\therefore V_{\text{Out}} = 0.9 \times V_{\text{In}} \quad \text{Eq 7}$$

As seen in the solutions (Equation 5 and Equation 7), the design requirements were achieved.

Plotting these solutions with a 1kHz square wave at V_{In} shows our expected behavior.

```
import matplotlib.pyplot as plt
import scipy.signal as signal
import numpy as np
t = np.linspace(0, 2, 1000)
Vi = signal.square(2*np.pi*t)
VoA = 0.5*Vi
VoB = 0.9*Vi
plt.plot(t, Vi)
plt.plot(t, VoA, '--')
plt.plot(t, VoB, '--')
plt.xlabel('Time (ms)')
plt.ylabel('Voltage (V)')
plt.title('Title')
plt.savefig('PartAWaveform.svg')
```

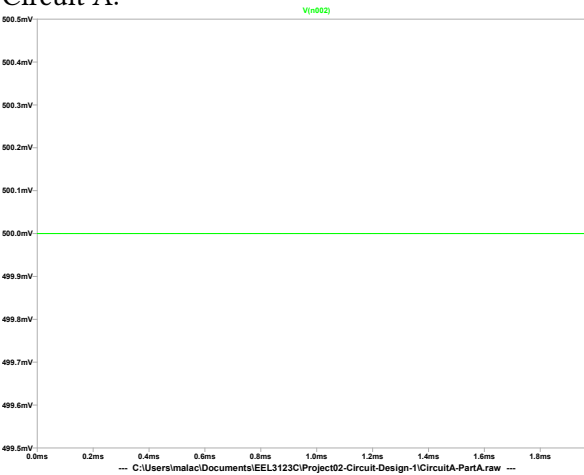


2.A.4 Simulation Results

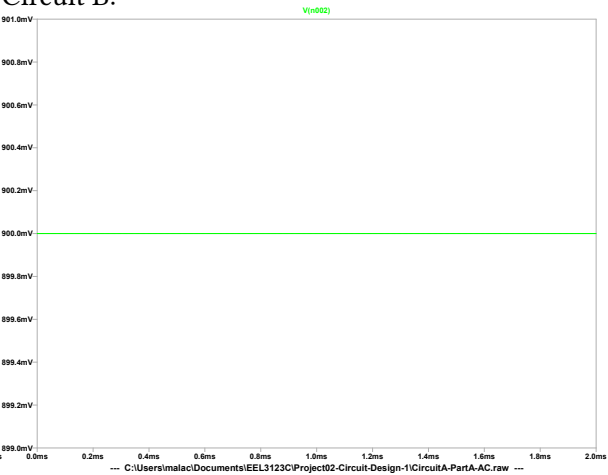
2.A.4.a DC Simulation

I set the input voltage to 1V and measured the output voltage at R_2 for the circuits in Figure 3 and Figure 4.

Circuit A:



Circuit B:



2.A.4.b AC Simulation

I set the input voltage to a 1V 1kHz square wave and measured the output voltage at R_2 for the circuits in Figure 3 and Figure 4.

Functions

☐ (none)

☒ PULSE(V1 V2 Tdelay Trise Tfall Ton Period Ncycles)

☐ SINE(Voffset Vamp Freq Td Theta Phi Ncycles)

☐ EXP(V1 V2 Td1 Tau1 Td2 Tau2)

☐ SFFM(Voff Vamp Fcar MD1 Fsig)

☐ PWL(t1 v1 t2 v2...)

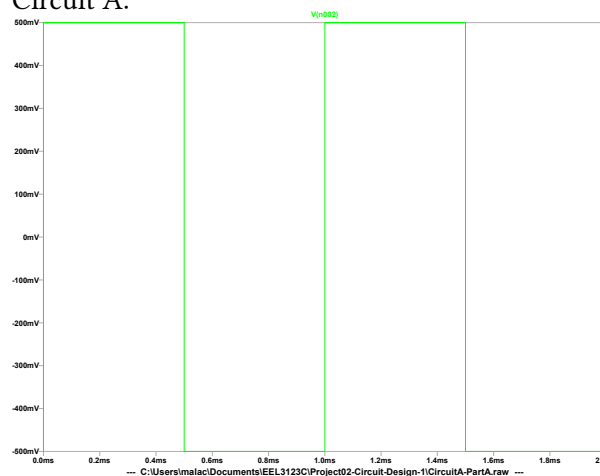
☐ PWL FILE:

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Von[V]:	1
Tdelay[s]:	0
Trise[s]:	0.0000000
Tfall[s]:	0.0000000
Ton[s]:	0.0005
Tperiod[s]:	0.001
Ncycles:	2

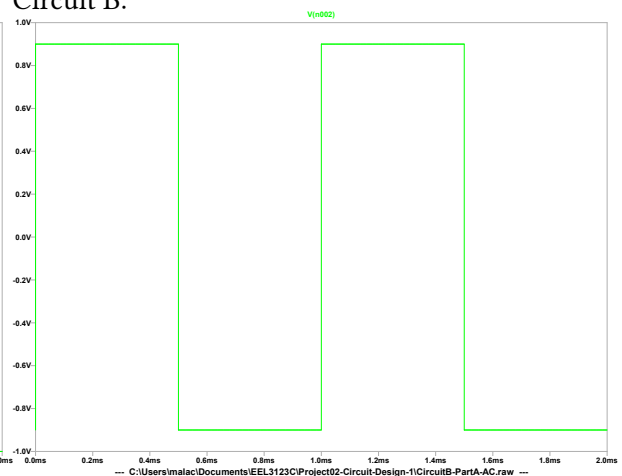
Smooth half-sine transitions ☐

Make this information visible on schematic: ☒

Circuit A:



Circuit B:



2.A.5 Experimental Results

2.A.6 Results Comparison

2.A.7 Conclusion

2.B Part B

2.B.1 Objective

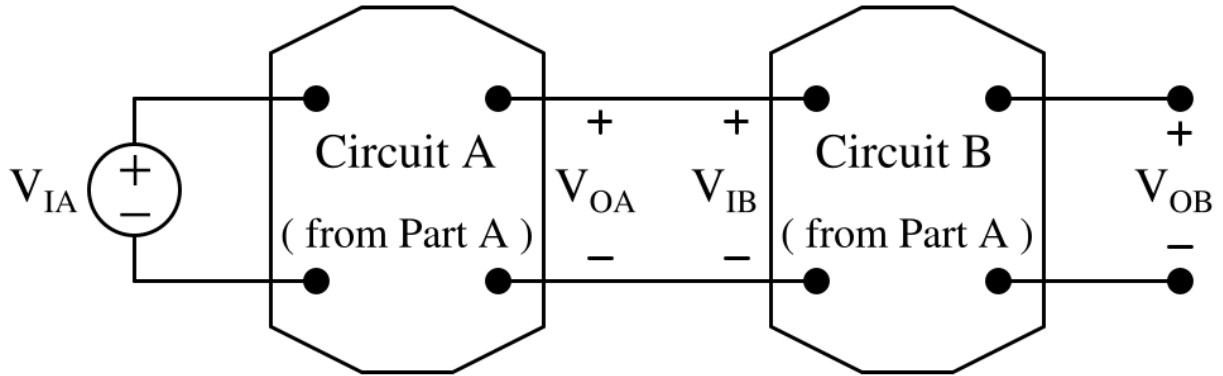


Figure 5: Cascaded circuits from part A

In this configuration, $V_{OB} \neq K_A \times K_B \times V_{IA}$. An analysis of the circuit in Figure 5 is shown below.

KCL@ $V_{R_{A2}}$

$$\frac{V_{In} - V_{R_{A2}}}{R_{A1}} = \frac{V_{R_{A2}}}{R_{A2}} + \frac{V_{R_{A2}} - V_{Out}}{R_{B1}} \quad \text{Eq 8}$$

KCL@ V_{Out}

$$\frac{V_{R_{A2}} - V_{Out}}{R_{B1}} = \frac{V_{Out}}{R_{B2}} \quad \text{Eq 9}$$

```
import sympy as sp
Vi, Vra2, Vo = sp.symbols('Vi Vra2 Vo')
eq1 = (Vi - Vra2)/1000 - (Vra2/9000 + (Vra2 - Vo)/1000)
eq2 = (Vra2 - Vo)/1000 - Vo/9000
print(sp.solve([eq1, eq2], Vra2, Vo))

>> {Vo: 81*Vi/109, Vra2: 90*Vi/109}
```

$$V_{Out} = \frac{81}{109} \times V_{In} \quad \text{Eq 10}$$

As this result shows, cascading the circuits does not simply multiply the gains of each individual circuit. This is due to the loading on circuit A by circuit B. This loading affects the voltage at $V_{R_{A2}}$, and consequentially, the output voltage aka $V_{R_{B2}}$.

2.B.2 Design

In order to correct this circuit, we must make sure that

$$\begin{aligned}V_{o_A} &= K_A \times V_{i_A} \\V_{o_B} &= K_B \times V_{i_B} \\V_{o_B} &= K_A \times K_B \times V_{i_A}\end{aligned}\quad \text{Eq 11}$$

where

$$\begin{aligned}K_A &= 0.5 \\K_B &= 0.9\end{aligned}\quad \text{Eq 12}$$

and using the same circuit analysis from Section 2.B.1, we find:

```
import sympy as sp
Vi, Vra2, Vo, Ra1, Ra2, Rb1, Rb2 = sp.symbols('Vi Vra2 Vo Ra1 Ra2 Rb1 Rb2')
eq1 = (Vi - Vra2)/Ra1 - (Vra2/Ra2 + (Vra2 - Vo)/Rb1)
eq2 = (Vra2 - Vo)/Rb1 - Vo/Rb2
eq3 = Vra2 - (0.5*Vi)
eq4 = Vo - (0.9*Vra2)
eq4 = Vo - (0.5*0.9*Vi)
sol = sp.solve([eq1, eq2, eq3, eq4], Vi, Vo, Vra2, Ra2, Rb2)
print(sol[0])
print("If Ra1 = 50k0hm and Rb1 = 10k0hm...")
print("Ra2 =")
print(sol[0][3].subs(Ra1,50_000).subs(Rb1,10_000))
print("Rb2 =")
print(sol[0][4].subs(Ra1,50_000).subs(Rb1,10_000))

>> (2.0*Vra2, 0.9*Vra2, Vra2, -10.0*Ra1*Rb1/(Ra1 - 10.0*Rb1), 9.0*Rb1)
>> If Ra1 = 50k0hm and Rb1 = 10k0hm...
>> Ra2 =
>> 100000.0000000000
>> Rb2 =
>> 90000.0000000000
```

This gives us the resistor values:

- $R_{A1} = 50k\Omega$
- $R_{A2} = 100k\Omega$
- $R_{B1} = 10k\Omega$
- $R_{B2} = 90k\Omega$

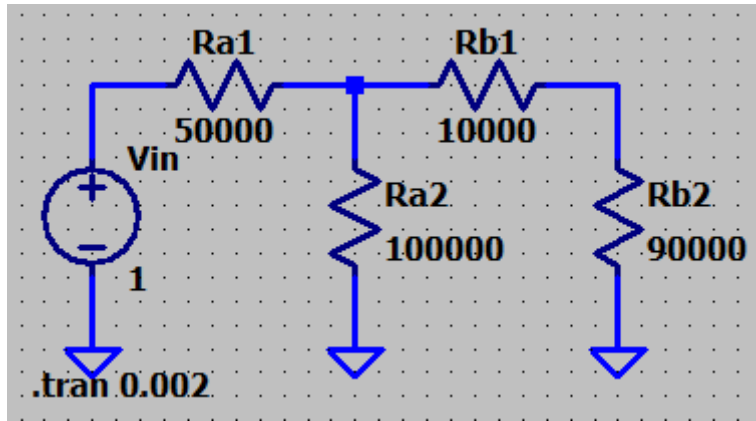
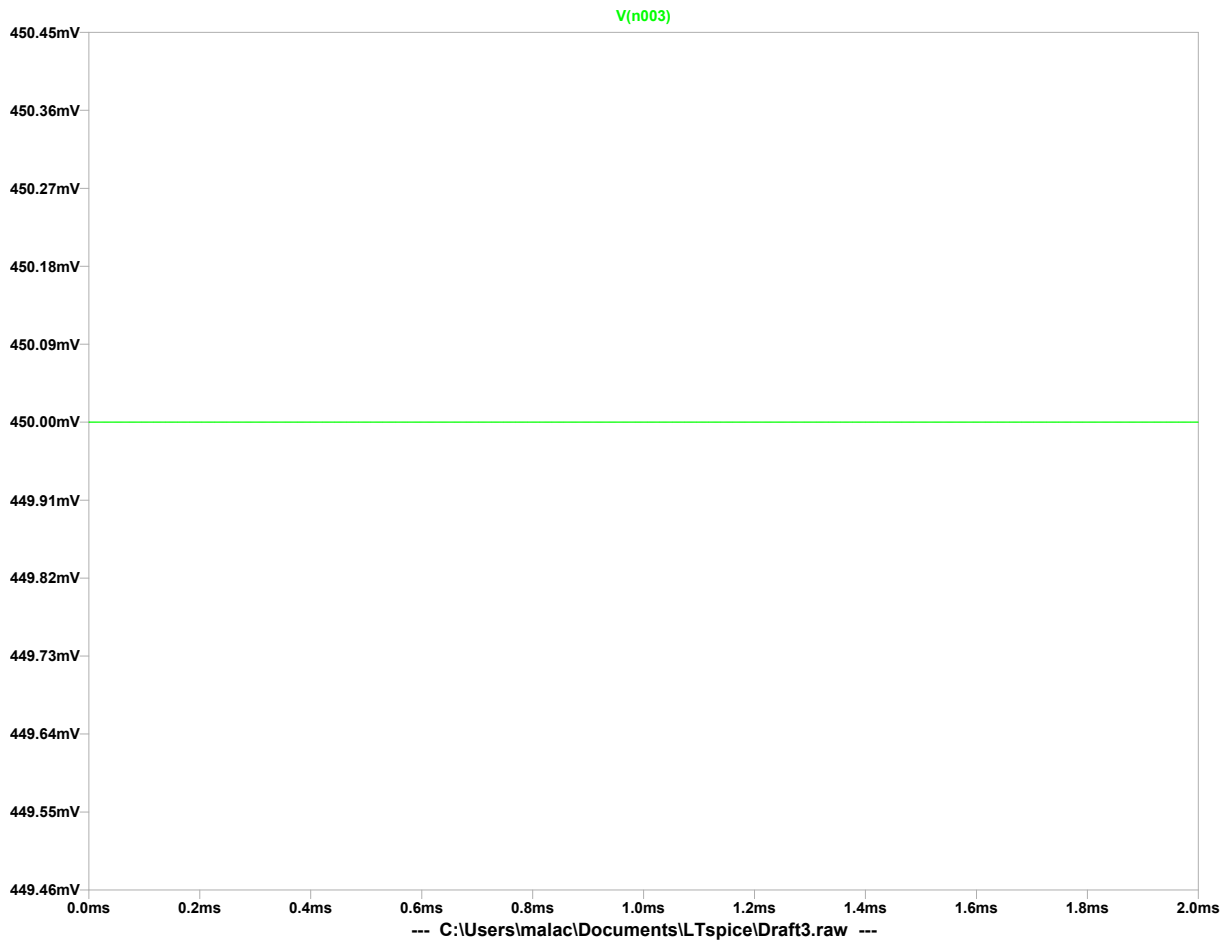


Figure 6: Updated Circuit A and B (cascaded)

2.B.3 Simulation Results

2.B.3.a DC Simulation

I set the input voltage to $1V$ and measured the output voltage at R_{B_2} for the circuit in Figure 6



2.B.3.b AC Simulation

I set the input voltage to a 1V 1kHz square wave and measured the output voltage at R_2 for the circuit in Figure 6

Functions

☐ (none)

☒ PULSE(V1 V2 Tdelay Trise Tfall Ton Period Ncycles)

☐ SINE(Voffset Vamp Freq Td Theta Phi Ncycles)

☐ EXP(V1 V2 Td1 Tau1 Td2 Tau2)

☐ SFFM(Voff Vamp Fcar MD1 Fsig)

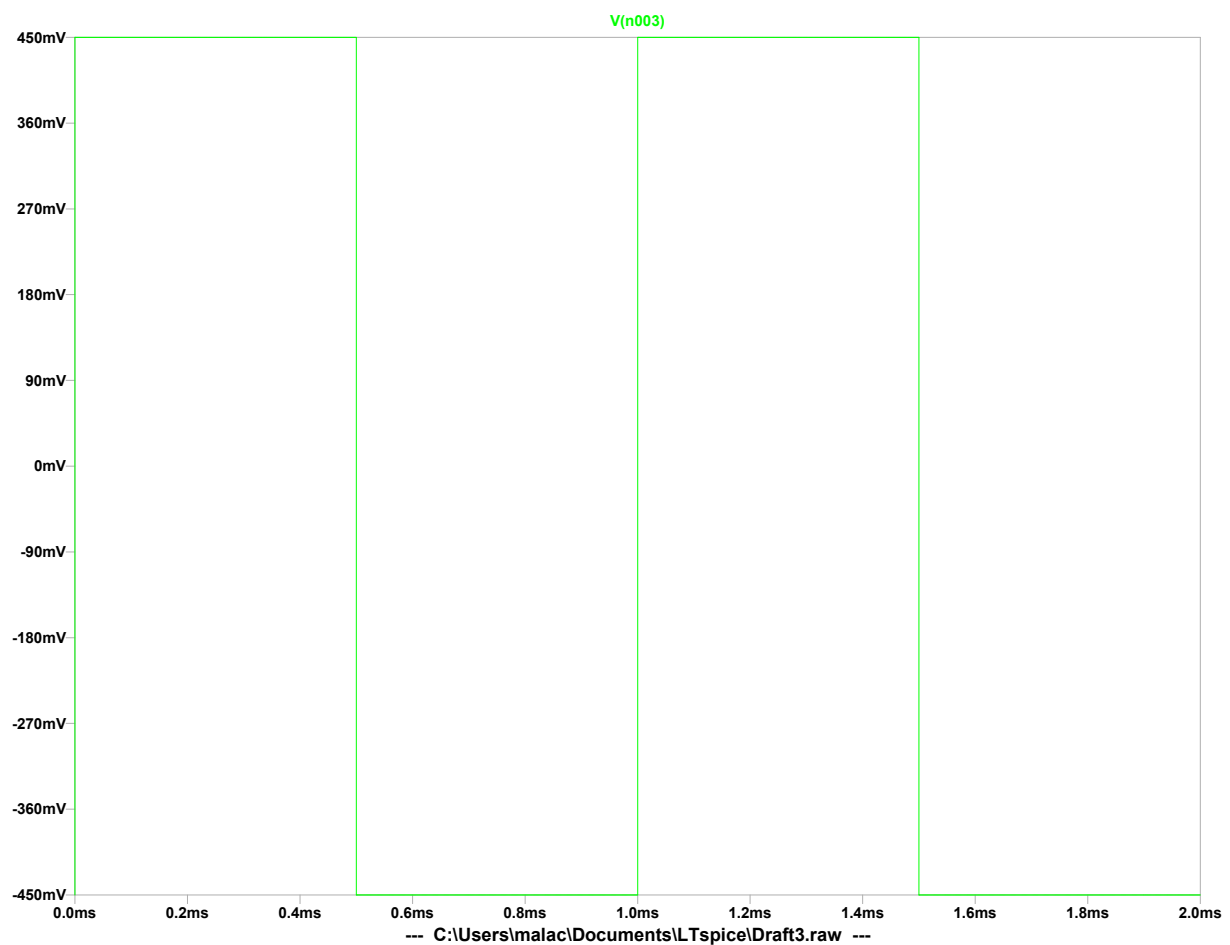
☐ PWL(t1 v1 t2 v2...)

☐ PWL FILE:

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Von[V]:	1
Tdelay[s]:	0
Trise[s]:	0.0000000
Tfall[s]:	0.0000000
Ton[s]:	0.0005
Tperiod[s]:	0.001
Ncycles:	2

Smooth half-sine transitions ☐

Make this information visible on schematic: ☒



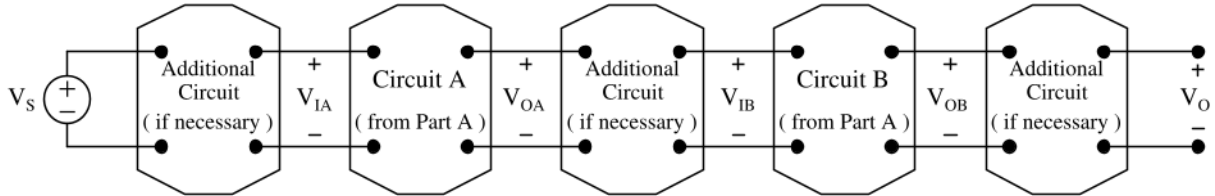
2.B.4 Experimental Results

2.B.5 Results Comparison

2.B.6 Conclusion

2.C Part C

2.C.1 Objective



Design Requirement:

- $V_{OA} = K_A \times V_{IA}$
- $V_{OB} = K_B \times V_{IB}$
- $V_{OB} = K_A \times K_B \times K_C \times V_{IA}$
- $K_A = 0.5$
- $K_B = 0.9$
- $K_C = 6$ (The cascaded circuit should be 180° out-of-phase too)

2.C.2 Design

To achieve the design requirement, we used a voltage follower (gain = 1) and an inverting amplifier (gain = -6). Originally we had:

$$\text{Circuit A} \rightarrow \text{Voltage Follower} \rightarrow \text{Circuit B} \rightarrow \text{Inverting Amp} \quad \text{Eq 13}$$

We then discovered that the inverting amplifier is loading Circuit B and affecting the output. So, we rearranged it as such:

$$\text{Circuit A} \rightarrow \text{Voltage Follower} \rightarrow \text{Inverting Amp} \rightarrow \text{Circuit B} \quad \text{Eq 14}$$

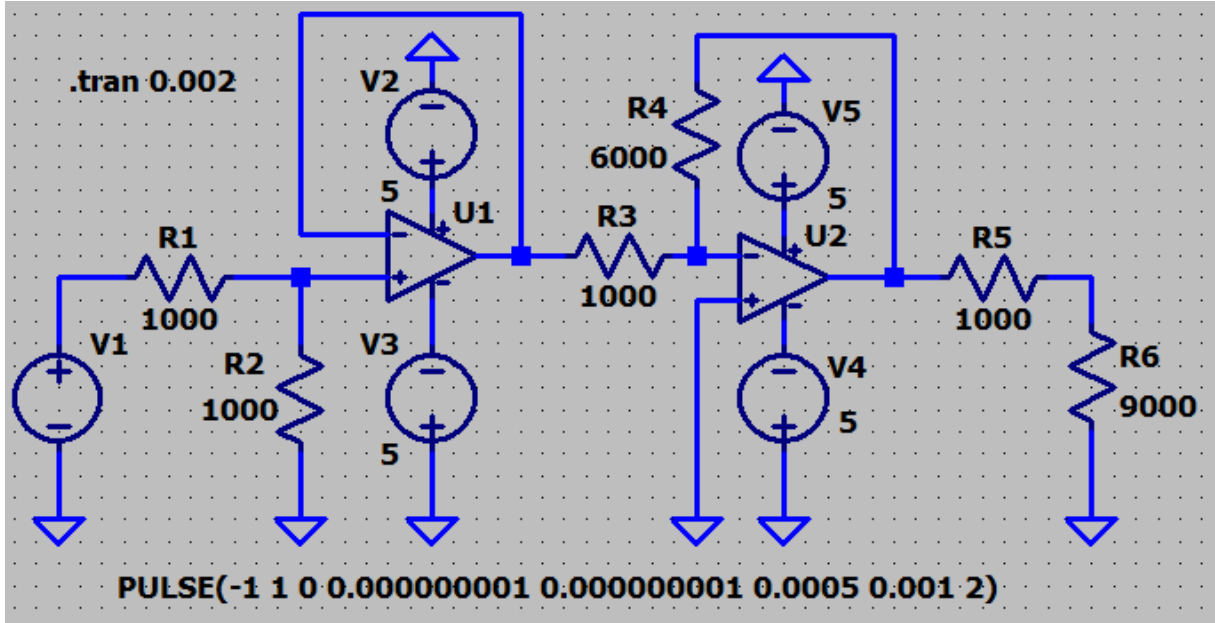


Figure 7: Updated Circuit A and B (cascaded) with voltage follower and inverting amplifier

2.C.3 Analysis Results

KCL@ $V_{R_{A2}}$

$$\frac{V_{In} - V_{R_{A2}}}{R_{A1}} = \frac{V_{R_{A2}}}{R_{A2}} + I_{V_{+1}}$$

Op Amp Current Behavior

Eq 15

$$I_{V_{+1}} = 0V$$

Op Amp Voltage Behavior

$$V_{+1} = V_{-1} = V_{O1}$$

KCL@ V_{+2}

$$\frac{V_{O1} - V_{-2}}{R_{In}} + \frac{V_{O2} - V_{-2}}{R_{Feedback}} = I_{V_{+2}}$$

Op Amp Current Behavior

Eq 16

$$I_{V_{+2}} = 0A$$

Op Amp Voltage Behavior

$$V_{+2} = V_{-2} = 0V$$

KCL@ V_{Out}

$$\frac{V_{O2} - V_{Out}}{R_{B1}} = \frac{V_{Out}}{R_{B2}}$$

Eq 17

```

import sympy as sp
Vi, VRA2, Vo2, Vout = sp.symbols('V_i V_R_A2 Vo_2 V_out')
eq10 = (Vi - VRA2)/1000 - (VRA2/1000 + 0)
eq11 = (VRA2 - 0)/1000 + (Vo2 - 0)/6000 - (0)
eq12 = (Vo2-Vout)/1000 - (Vout/9000)
print(sp.solve([eq10, eq11, eq12],VRA2, Vo2, Vout))

>> {V_R_A2: V_i/2, V_out: -27*V_i/10, Vo_2: -3*V_i}

```

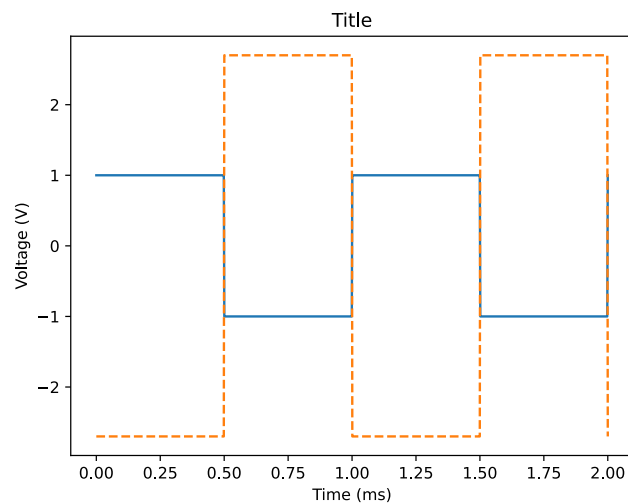
$$\begin{aligned}
 V_{\text{Out}} &= -\frac{27}{10} \times V_{\text{In}} \\
 &= 0.5 \times 0.9 \times 6 \times -1 \times V_{\text{In}}
 \end{aligned}
 \tag{Eq 18}$$

Plotting these solutions with a 1kHz square wave at V_{In} shows our expected behavior.

```

import matplotlib.pyplot as plt
import scipy.signal as signal
import numpy as np
t = np.linspace(0,2, 1000)
Vi = signal.square(2*np.pi*t)
Vout = 0.5*0.9*6*-1*Vi
plt.plot(t,Vi)
plt.plot(t,Vout,"--")
plt.xlabel('Time (ms)')
plt.ylabel('Voltage (V)')
plt.title('Title')
plt.savefig('PartCWaveform.svg')

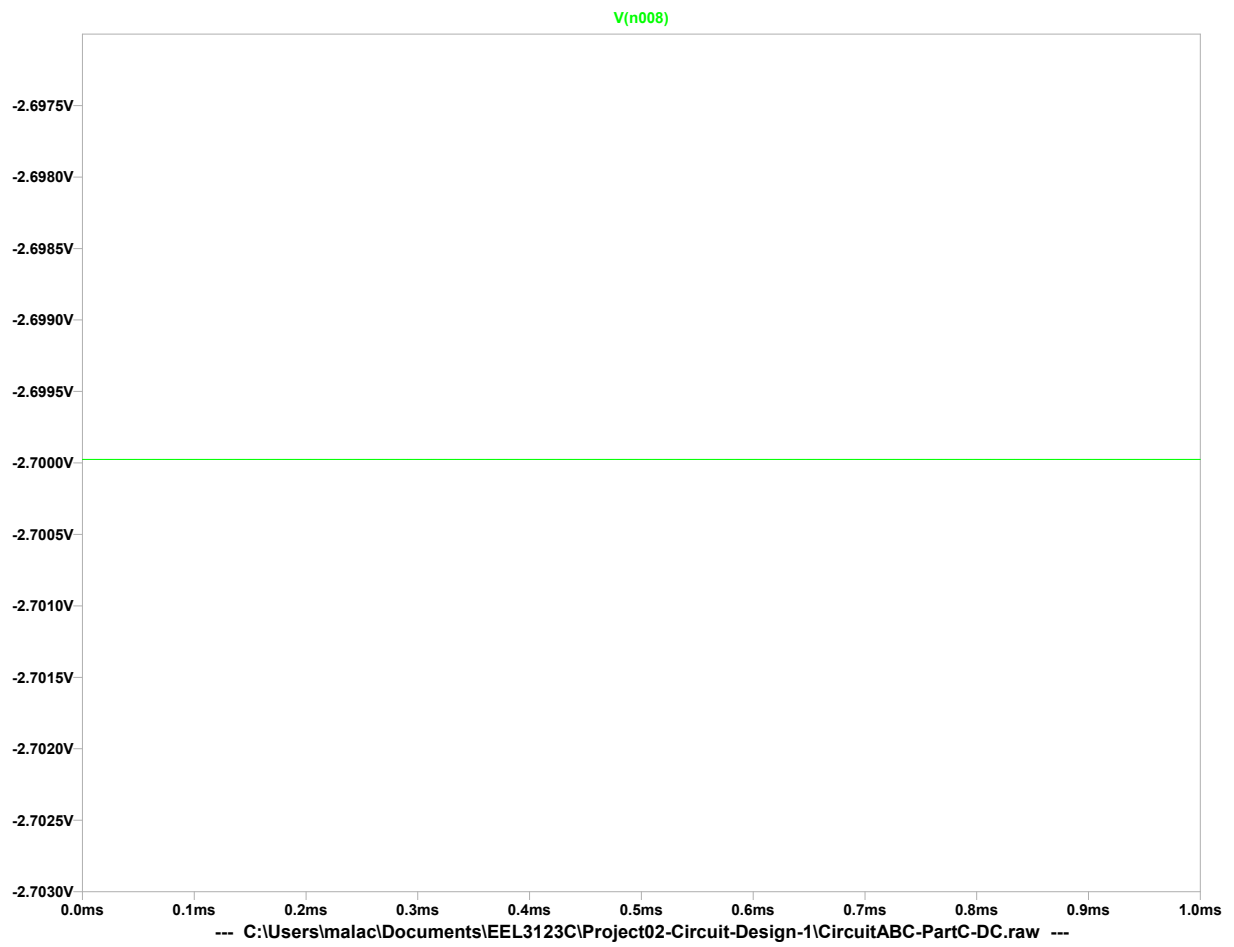
```



2.C.4 Simulation Results

2.C.4.a DC Simulation

I set the input voltage to 1V and measured the output voltage at R_{B_2} for the circuit in Figure 7



2.C.4.b AC Simulation

I set the input voltage to a 1V 1kHz square wave and measured the output voltage at R_2 for the circuit in Figure 7

Functions

☐ (none)

☒ PULSE(V1 V2 Tdelay Trise Tfall Ton Period Ncycles)

☐ SINE(Voffset Vamp Freq Td Theta Phi Ncycles)

☐ EXP(V1 V2 Td1 Tau1 Td2 Tau2)

☐ SFFM(Voff Vamp Fcar MD1 Fsig)

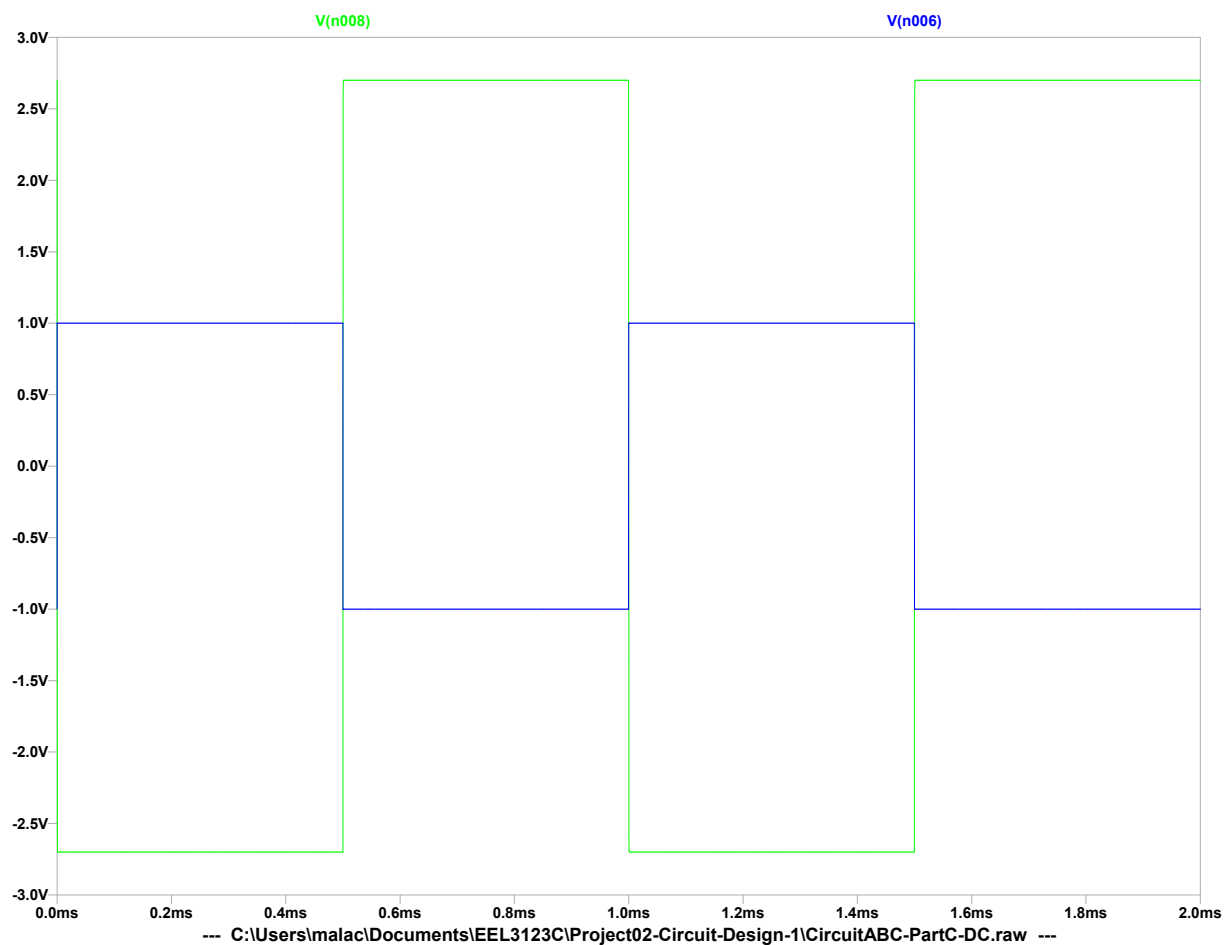
☐ PWL(t1 v1 t2 v2...)

☐ PWL FILE:

Vinitial[V]:	-1
Von[V]:	1
Tdelay[s]:	0
Trise[s]:	0.0000000
Tfall[s]:	0.0000000
Ton[s]:	0.0005
Tperiod[s]:	0.001
Ncycles:	2

Smooth half-sine transitions ☐

Make this information visible on schematic: ☒



2.C.5 Experimental Results

2.C.6 Results Comparison

2.C.7 Conclusion